The Binomial Theorem

Section 1: Pascal's Triangle and Combinations

1 State the formula for the combination $\binom{n}{r}$.

a. $\frac{n!}{r!\,(n-r)!}$

b. $\frac{n!}{(n-r)!}$

c. $\frac{n!}{r!}$

d. $n \times r$

2 Evaluate $\binom{6}{2}$.

a. 15

b. 30

c. 12

d. 20

3 What is row 4 of Pascal's Triangle (corresponding to $n = 4$)?

a. $1, 4, 6, 4, 1$

b. $1, 3, 3, 1$

c. $1, 5, 10, 10, 5, 1$

d. $1, 4, 8, 4, 1$

Section 2: Expanding Binomials

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4 State the Binomial Theorem expansion for $(a+b)^n$.

a. $\sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$

b. $\sum_{r=0}^{n} a^{n-r} b^r$

c. $\sum_{r=1}^{n} \binom{n}{r} a^r b^{n-r}$

d. $\sum_{r=0}^{n} n!\, a^r b^r$

5 Expand $(x+2)^4$.

a. $x^4 + 8x^3 + 24x^2 + 32x + 16$

b. $x^4 + 4x^3 + 6x^2 + 4x + 16$

c. $x^4 + 16$

d. $x^4 + 8x^3 + 16x^2 + 32x + 16$

6 Find the first three terms in the expansion of $(2x-1)^5$ in descending powers of x.

a. $32x^5 - 80x^4 + 80x^3$

b. $32x^5 - 16x^4 + 8x^3$

c. $32x^5 + 80x^4 + 80x^3$

d. $32x^5 - 40x^4 + 20x^3$

Section 3: Finding Specific Terms

7 State the formula for the $(r+1)$-th term in the expansion of $(a+b)^n$.

a. $T_{r+1} = \binom{n}{r} a^{n-r} b^r$

b. $T_{r+1} = \binom{n}{r+1} a^{n-r} b^r$

c. $T_{r+1} = \binom{n}{r} a^r b^{n-r}$

d. $T_{r+1} = a^{n-r} b^r$

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8 Find the 4th term in the expansion of $(x+3)^7$.

a. $945x^4$

b. $315x^4$

c. $2835x^3$

d. $105x^5$

9 Find the coefficient of x^3 in the expansion of $(2x-3)^6$.

a. -4320

b. 4320

c. -1440

d. 2160

Section 4: Term Independent of x and Sum of Coefficients

10 Find the term independent of x (the constant term) in the expansion of $\left(x+\frac{2}{x}\right)^6$.

a. 160

b. 80

c. 64

d. 20

11 What is the sum of all coefficients in the expansion of $(2x-3y)^4$?

a. 1

b. -1

c. 625

d. 81

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12 What is the sum of the elements in row n of Pascal's Triangle (i.e. $\sum_{r=0}^{n} \binom{n}{r}$)?

a. 2^n

b. n^2

c. $2n$

d. 2^{n-1}

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